Module 1

Supply and Demand

Unit 1: Demand

Objectives of the Study:

After studying this unit, the student should be able to:

- understand the concept of demand curves
- define demand and its determinants
- explain the Law of Demand
- identify differences between Individual and Market Demand Curve

Demand

In economics, demand refers to the various quantities of a good or service that people will be and able to purchase at various prices during a period of time. It is important to note that a mere desire for a good or service does not constitute demand. Demand implies both the desire to purchase and ability to pay for the good. Unless demand is backed by purchasing power, it does not constitute demand. Further, demand does not refer to the specific quantity that will be purchased at some particular price, but refer to a series of quantities and their associated prices.

Demand for a commodity refers to the desire for the commodity backed by willingness and ability to pay for that commodity. So, demand includes the desire to buy the commodity accompanied by the willingness to buy it and sufficient purchasing power to purchase it. For instance-Everyone might have willingness to buy Car but only a few have the ability to pay for it. Thus, everyone cannot be said to have a demand for the car.

Demand may arise from individuals, household and market. When goods are demanded by individuals, it is called as individual demand. Goods demanded by household constitute household demand. Demand for a commodity by all individuals/households in the market in total constitutes market demand. The Market demand is the horizontal summation of individual demand.

Law of Demand

Law of demand expresses the functional relationship between price and quantity demanded. According to the law of demand, other things being equal, if the price of the commodity falls the quantity demanded of it will rise and if the price of the commodity rises, its quantity demanded will decline. Thus, according to law of demand, there is an inverse relationship between price and quantity demanded, other things remaining the same. The other things which are assumed to be constant are tastes and preferences of the consumer, the income of the consumer, prices of related commodities etc. Thus, the law of demand assumes that all things other than price remain constant.

The law of demand states that there is an inverse relationship between quantity demanded of a commodity and its price, other factors being constant. In other words, higher the price, lower the demand and vice versa, other things remaining constant.

Symbolically,

$$\mathbf{D_X = f\ (P_X)}, \text{ ceteris paribus}$$

where,

D_X = Quantity demanded of good X

P_X = Price of the good X

The law of demand states that if remaining things are constant then as price of a commodity increases demand for the commodity decreases and as price of a commodity decreases demand for the commodity increases. Suppose a consumer is willing to buy 100 units of a commodity when its price is ₹10 per unit. As price of the commodity increased to ₹15 per unit then consumer is willing to buy 80 units of the commodity and vice versa. It shows definitely inverse relation between price and demand for the commodity.

Assumptions of the law of demand

The law is valid only when the following assumptions hold:

(a) The price of the related goods remains the same.

(b) The income of the consumers remains unchanged.

(c) Tastes and preferences of the consumers remain the same.

(d) All the units of the goods are homogeneous.

(e) Commodity should be a normal good.

Demand Schedule

Demand schedule is a tabular representation of the quantity demanded of a commodity at various prices. That is demand schedule is a tabular presentation showing the different quantities of a good that buyers of the good are willing to buy at different prices during a given period of time. Table 1.1 shows a hypothetical demand schedule for wheat.

Table 1.1 Demand Schedule for Wheat

Price (₹per kg)	Quantity Demanded (kg per month)	Reference Point
20	6	A
30	5	B
40	4	C
50	3	F

The demand schedule shows an inverse relationship between price and the quantity demanded. The consumer is willing to pay ₹50 per kg to buy 3 kg of wheat each month. If the price reduces to ₹40 per kg, he would be willing to buy an additional one kg per month and so on. This implies that lower the price more will be the demand and vice-versa.

The Demand Curve

Demand curve is a diagrammatic representation of demand schedule. It is a graphical representation of price- quantity relationship. In Fig. 1.1 demand curve for wheat is drawn which shows different quantities of wheat demanded at different prices in a month.

The demand curve, d, slopes downward to the right or is negatively sloped. This law of downward sloping demand has been empirically tested and verified. The independent variable (price) is measured along the y-axis and dependent variable (quantity) is measured along the x-axis. The demand curve shows the quantity demanded by the consumers at each price.

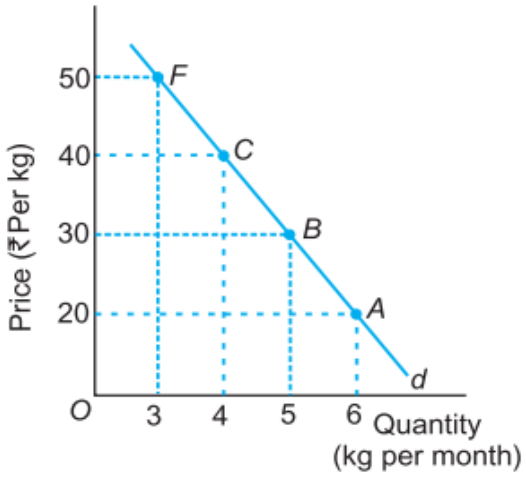


Fig. 1.1 The Demand Curve

Demand curve has a negative slope, i.e., it slopes downwards from left to right depicting that with increase in price, quantity demanded falls and vice versa. The reasons for a downward sloping demand curve can be explained as follows:

1. Income effect- With the fall in price of a commodity, the purchasing power of consumer increases. Thus, he can buy same quantity of commodity with less money or he can purchase greater quantities of same commodity with same money. Similarly, if the price of a commodity rises, it is equivalent to decrease in income of the consumer as now he has to spend more for buying the same quantity as before. This change in purchasing power (real income) due to price change is known as income effect.

2. Substitution effect- When price of a commodity falls, it becomes relatively cheaper compared to other commodities whose price have not changed. Thus, the consumer tends to consume more of the commodity whose price has fallen, i.e., they tend to substitute that commodity for other commodities which have not become relatively dear.

3. Law of diminishing marginal utility- It is the basic cause of the law of demand. The law of diminishing marginal utility states that as an individual consumes more and more units of a commodity, the utility derived from it goes on decreasing. So as to get maximum satisfaction, an individual purchase in such a manner that the marginal utility of the commodity is equal to the price of the commodity. When the price of commodity falls, a rational consumer purchases more so as to equate the marginal utility and the price level. Thus, if a consumer wants to

purchase larger quantities, then the price must be lowered. This is what the law of demand also states.

Exceptions to Law of Demand

The instances where law of demand is not applicable are as follows-

1. There are certain goods which are purchased mainly for their snob appeal, such as, diamonds, air conditioners, luxury cars, antique paintings, etc. These goods are used as status symbols to display one's wealth. The more expensive these goods become, more valuable will be they as status symbols and more will be there demand. Thus, such goods are purchased more at higher price and are purchased less at lower prices. Such goods are called as conspicuous goods.

2. The law of demand is also not applicable in case of Giffen goods. Giffen goods are those inferior goods, whose income effect is stronger than substitution effect. These are consumed by poor households as a necessity. For instance, potatoes, animal fat oil, low quality rice, etc. An increase in price of such good increases its demand and a decrease in price of such good decreases its demand.

3. The law of demand does not apply in case of expectations of change in price of the commodity, i.e., in case of speculation. Consumers tend to purchase less or tend to postpone the purchase if they expect a fall in price of commodity in future. Similarly, they tend to purchase more at high price expecting the prices to increase in future.

The Demand Function

Demand function is showing relationship between the quantity demanded of a commodity and the factors influencing demand.

$$\mathbf{D_x = f\ (P_x, P_y, T, Y, A, N, Ep, U)}$$

In the above equation,

D_x = Quantity demanded of a commodity

P_x = Price of the commodity

P_y = Price of related goods

T = Tastes and preferences of consumer

Y = Income level

A = Advertising and promotional activities

N = No. of consumers (Size of the market)

Ep = Consumer's expectations about future prices

U = Specific factors affecting demand for a commodity such as seasonal changes, taxation policy, availability of credit facilities, etc.

Factors Affecting Individual Demand for a Good

There are following factors which affect demand for a commodity

1. **Price of the Commodity**

There is inverse relationship between price of a commodity and demand for a commodity. In general, demand for a commodity is more at lower price and less at a higher price and vice versa. But this relationship does not exist in Giffen goods. In case of Giffen goods there is direct relationship between price and demand

2. **Prices of Other Goods**

Demand for good x is influenced by the prices of other good (z). It is called cross price demand.

$$D_X = f(P_Z), \text{ ceteris paribus}$$

The relationship depends upon the relation between two goods x and z. Two situations can arise:

(a) When X and Z are Substitutes

Substitute goods are those which are an alternative to one another in consumption. They satisfy same human want with equal ease. Examples are: tea or coffee; wheat or rice; ink pen or ball pen; a Maruti car or a Zen car, Pepsi or Coca Cola, Lux Supreme or Lifebuoy Gold. This substitute relationship arises because the goods have a similar technology or have a similar price.

Example.

If the price of tea rises from ₹200 to ₹250 per kg it would cause an increase in demand for coffee from C_1 units to C_2 units at price OP_1 . Fig. 1.2 illustrates a rightward shift in the demand curve to coffee from d to d_1 when the goods tea and coffee are substitutes. An increase in the price of a substitute good increases the quantity demanded of the other good. If there is an increase in the price of a substitute good, the demand curve shifts rightward.

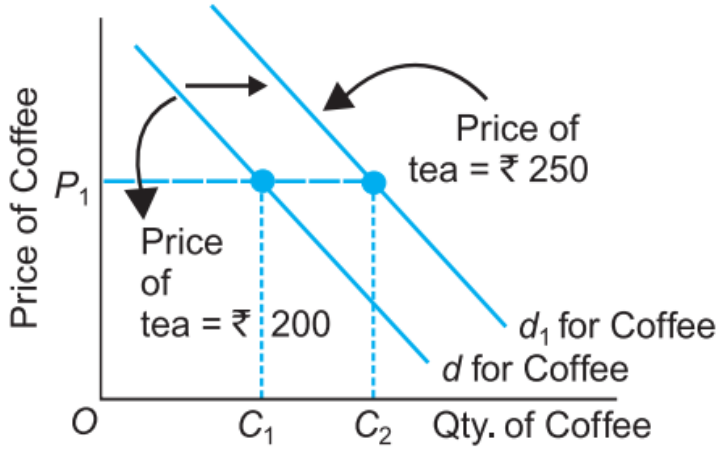


Fig. 1.2 Demand Curve when X and Y are Substitutes

Thus, demand for a good usually moves in the same direction to a change in price of its substitutes.

(b) When X and Z are Complements

Complementary goods are those which are jointly used or consumed together to satisfy a want. Examples are: tea and sugar; car and petrol; pen and ink; bread and butter; cigarettes and cigarette lighter; compact disc player and compact discs. Example. If price of petrol rises from ₹35 to ₹40 a litre, then quantity demanded of car will reduce from C_1 to C_2 units at price of rate OP_1 , other things remaining the same. This is graphically shown in Fig. 1.3 There is a leftward shift of the demand curve of car from d to d_1 when the two goods are complementary. That is, if there is increase in the price of complementary good, the demand curve shifts leftward.

Thus, demand for a good move in the opposite direction to a change is price of its complementary good.

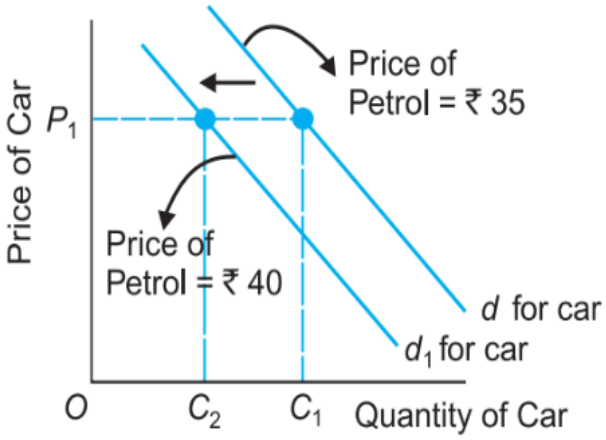


Fig. 1.3 Demand Curve when X and Z are Complements

3) **Income of the Consumer**

Changes in money of the consumer changes the budget constraint facing the consumer, causing him to change his demand for goods. It is called income demand. Symbolically,

$$D_X = f(Y), \text{ ceteris paribus}$$

How a change in the income will affect the demand for a good depends upon the type of the good:

(a) If x is a normal good then with an increase in income, consumer buys more of the good. Goods whose demand rises when income rises are called normal goods. Example: clothes, books, etc.

(b) If x is an inferior good then an increase in income causes its demand to decrease. This is because as income rises, purchasing power rises and consumers substitute more superior goods for inferior goods. Goods whose demand falls when income rises are called inferior goods. Example: Coarse cereals.

Graphically, the relationship between quantity demanded of good x and income of the consumer is shown as in Fig 1.4

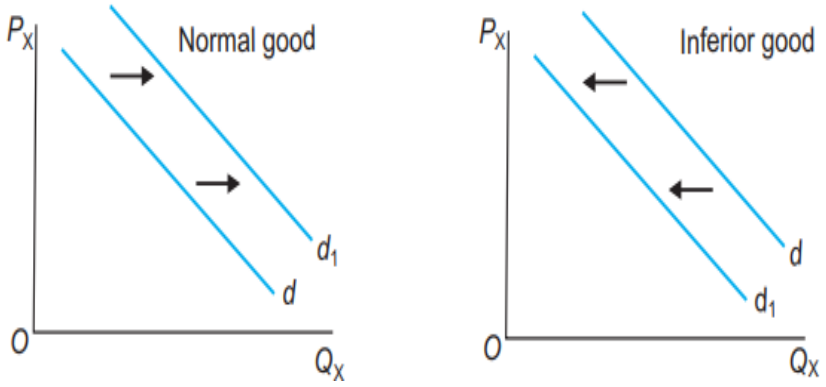


Fig 1.4 Effect of Rise in Income on Demand

The figure shows that in case of normal goods as income rises, demand increases and in case of inferior goods as income rises, demand decreases.

4) **Consumer's Tastes and Preferences**

Any change in consumer's tastes causes demand to change. If there is a change in tastes in favour of a good, then it will lead to increase in demand and any unfavourable change will lead to decrease in demand. The relationship is shown graphically by shifts of demand curve in Fig 1.5.

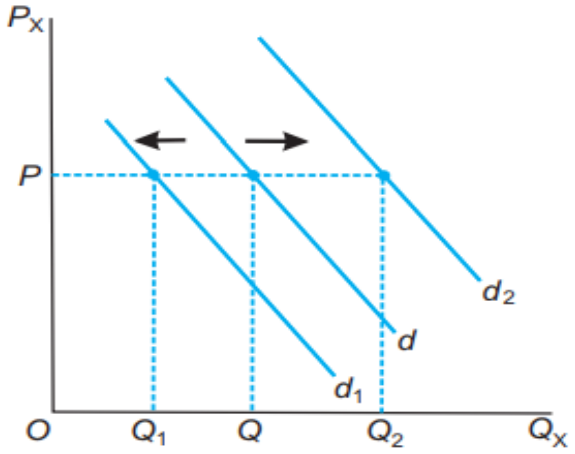


Fig 1.5 Shift of Demand Curve: Due to change in tastes and preferences

Increased preference for a good is shown by increase in demand, i.e., rightward shift of demand curve from d to d_2. It shows that more is demanded at each price. At price OP, the consumer will now demand a larger quantity OQ_2 compared to OQ (OQ is the amount demanded before the change in taste). Decreased preference for a good is shown by decrease in demand, i.e.,

leftward shift of demand curve from d to d_1. It shows that less is demanded at each price. At price OP, the consumer will now demand a smaller quantity OQ_1 compared to OQ.

5) **Future Expectations of Buyers**

Future expectation is also one of the factors which causes change in demand. If it is expected by the consumer that the price of the commodity will rise in future, he will start buying more units of the commodity in the present, at the existing price. Similarly, if he expects that price will fall in future, he will buy less quantity of the commodity in the present, even if the price, in present is less than the price in past. The relationship is shown graphically by shifts of demand curve in Fig. 1.6.

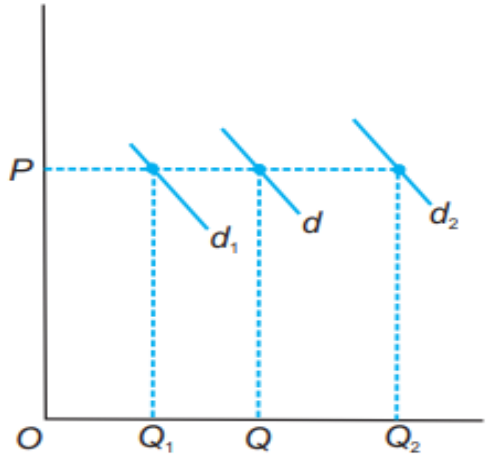


Fig. 1.6 Shift of Demand Curve: Due to change in expectations of buyers

The figure shows that if consumers expect rise in future price of the commodity they buy more units in present (Q to Q_2) at existing price (P) and demand curve shifts to the right (d to d_2). But if it is expected that price will fall in future, they buy less units in present (Q to Q_1) at existing price (P) and the demand curve shifts to the left (d to d_1).

Summing Demand Curves

Although the behaviour of an individual in respect of selection and purchase of goods forms the basis of demand theory, the aggregate demand or market demand for a good is most important for its producer. The aggregate quantity of a good that the buyers purchase or demand at a particular price and in a particular period is called the market demand for the good at the

said price. Also, the curve that gives us the market demand for a good at any particular price is known as its market demand curve.

It is obvious from the definition of market demand that the horizontal or lateral summation of the individual demand curves for a good would give us its market demand curve. The market demand curve for a good would also slope downward towards right, since, owing to the law of demand. Let there be two households A and B in the market for wheat. By aggregating or summing their individual demands, market demand is obtained. It is shown in Table 1.2.

Table 1.2 Individual and Market Demand Schedule

Price (₹ per kg)	Individual Demand Schedule for wheat (kg per month)		Market Demand Schedule (kg per month) $Q_A + Q_B$
	Q_A	Q_B	
2	4	5	9
4	3	4	7
6	2	3	5
8	1	2	3

Graphically, the market demand curve is a horizontal summation of the two individual demand curves. The derivation of the market demand curve is shown in Fig. 1.7.

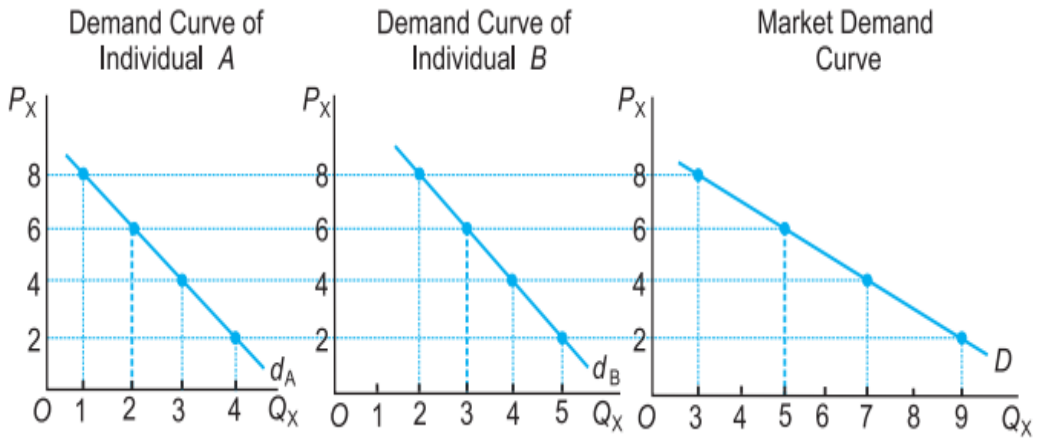


Fig. 1.7 Individual Demand Curves and Market Demand Curve

where

d_A and d_B = Individual demand curves for two consumers A and B.

D = Market demand curve. It is the lateral or horizontal summation of d_A and d_B curves at each and every price. It obeys the law of downward sloping demand.

Change in Quantity demanded (movement along the demand curve) and Change in Demand Curve (shift in demand curve)

1) Shift in Demand Curve

The demand curve shift to the right or left of the original demand curve parallelly when we reflex the assumption of other things remaining constant. In a shift, a new demand curve is drawn. A shift of the demand curve can bring about:

(a) Increase in demand, or

(b) Decrease in demand.

a) Increase in Demand.

It refers to more demand at a given price. The causes of increase in demand are:

(i) Increase in the income of the consumers in case of normal goods.
(ii) Decrease in the income of the consumers in case of inferior goods.
(iii) Increase in the price of substitute goods.
(iv) Fall in the price of complementary goods.
(v) Consumers' taste becoming stronger in favour of the good.

Increase in demand can be shown with a demand schedule given in Table 1.3

Table 1.3 Increase in Demand

Price (₹)	Quantity (Units)
3	90
3	100

The shift in demand curve has been shown graphically as in Fig. 1.8.

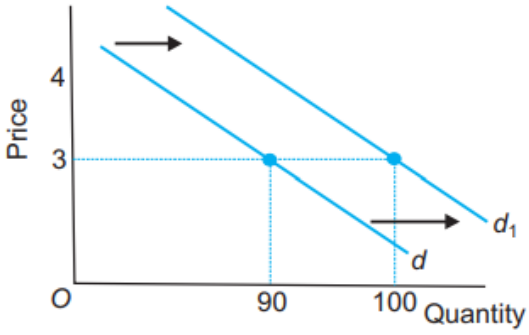


Fig. 1.8 Shift in Demand Curve: Increase in Demand

In the figure, d is the original demand curve. An increase in demand is shown by rightward shift of the demand curve from d to d_1. An increase in quantity demanded shows that at original price of ₹3, more units (100 units) of the good are demanded. In the original situation 90 units were demanded.

(a) Decrease in Demand.

It refers to less demand at the given price. It occurs due to unfavourable changes in factors other than the price of the good. The causes of decrease in demand are:

(i) Fall in the income of the consumers in case of normal goods.

(ii) Rise in the income of the consumers in case of inferior goods.

(iii) Fall in the price of substitute goods.

(iv) Rise in the price of complementary goods.

(vi) Consumers' taste becoming unfavourable towards the good.

Decrease in demand can be shown with the help of a demand schedule given in Table 1.4

Table 1.4 Decrease in Demand

Price (₹)	Quantity (Units)
3	90
3	80

The shift in demand curve has been shown graphically as in Fig. 1.9.

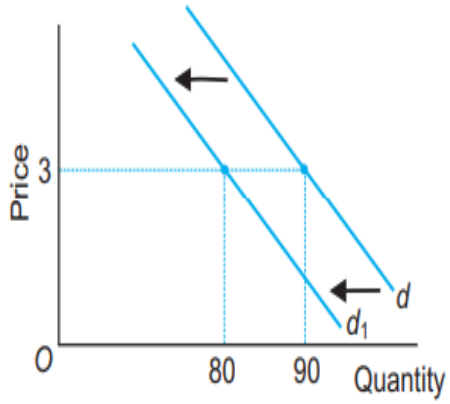


Fig. 1.9 Shift in Demand Curve: Decrease in Demand

In the figure, d is the original demand curve. A decrease in demand is shown by a leftward shift of the demand curve from d to d_1. A decrease in demand shows that at the original price of ₹3, lesser units (80 units) of the good are demanded. In the original situation 90 units were demanded. Table 1.5 summarises the difference in the causes of shift in the demand.

Table 1.5 Difference in the Causes of Shift in the Demand Curve

Increase in Demand (Upward or rightward shift in demand)	Decrease in Demand (Downward or leftward shift in demand)
Increase in the income of the consumers in case of normal goods.	Decrease in the income of the consumers.
Decrease in the income of the consumer in case of inferior goods.	Increase in the income of the consumer in case of inferior goods.
Increase in the price of substitute goods.	Decrease in the price of substitute goods.
Fall in the price of complementary goods.	Rise in the price of complementary goods.
Changes in tastes and preferences in favour of a commodity.	Changes in tastes and preferences against a commodity.

2) Movement along the Demand curve

A movement along the demand curve is caused by a change in the price of the good, other things remaining constant. It is also called change in quantity demanded of the commodity. Movement is always along the same demand curve, i.e., no new demand curve is drawn. Movement along a demand curve can bring about:

(a) Expansion of demand, or (b) Contraction of demand

Expansion or Extension of demand refers to rise in demand due to fall in the price of the good. It is represented by a downward movement along the same demand curve. It is also known as an increase in quantity demanded. On the other hand, Contraction of demand refers to fall in demand due to rise in the price of the good. It is represented by an upward movement along the same demand curve. It is also known as a decrease in quantity demanded. Extension or contraction of demand can be shown in Table 1.6.

Table 1.6 Expansion and Contraction

Price (₹)	Quantity (Units)	Point in the Graph
2	100	C (expansion)
3	90	A (original)
4	80	B (contraction)

The original point is A where the quantity demanded is 90 at price ₹3. When the price increases to ₹4 quantity demanded will decrease into 80 units. The movement from 90 to 80 is the contraction of demand and is represented by point B. In contrast when price decreases to ₹2, quantity demanded will increase into 100. This is called expansion of demand and is represented by point C. Movement along a demand curve is graphically illustrated in Fig. 1.10.

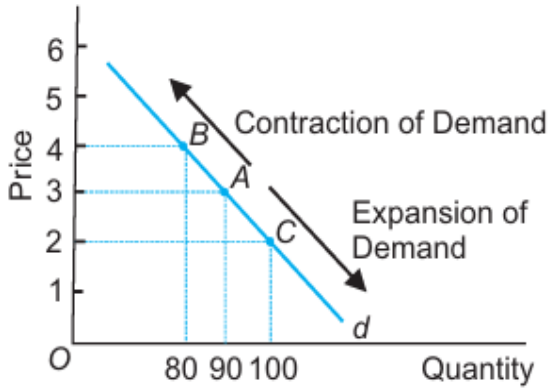


Fig. 1.10. Movement along Demand Curve

Point A on the demand curve d, is the original situation. An upward movement from point A to a point such as point B shows contraction or lesser quantity demanded at a higher price. A downward movement from point A to a point such as point C shows expansion or more quantity demanded at a lower price.

Multiple Choice Questions

1. What kind of relationship exists between demand for a good and price of its substitute goods?

(a) Direct (b) Inverse

(c) No effect (d) Can be direct or inverse

Answer: (a)

2. What kind of relationship exists between price of a good and demand of its complementary good?

(a) Direct (b) Inverse

(c) No effect (d) Can be direct or inverse

Answer: (b)

3. What kind of relationship exists between income and demand of inferior good?

(a) Direct (b) Inverse

(c) No effect (d) Can be direct or inverse

Answer: (b)

4. What kind of relationship exists between income and demand of a normal good?

(a) Direct (b) Inverse

(c) No effect (d) Can be direct or inverse

Answer: (a)

5. How is law of demand expressed functionally?

(a) $D_X = f(P_X)$, ceteris paribus (b) $D_X = f(P_Z)$, ceteris paribus

(c) $D_X = f(Y)$, ceteris paribus (d) $D_X = f(T)$, ceteris paribus

Answer: (a)

6. When demand curve is downward sloping, its slope is:

(a) Negative (b) Positive

(c) Constant (d) Zero

Answer: (a)

7. Income effect states that as price of a good falls, demand rises because there is rise in:

(a) Money income (b) Real income

(c) Relative price of other goods (d) Marginal utility

Answer: (b)

8. Substitution effect states that as price of a good falls, demand rises because there is rise in:

(a) Money income (b) Real income

(c) Relative price of other goods (d) Marginal utility

Answer: (c)

9. Giffen good is:

(a) An inferior good (b) One with high negative income elasticity of demand

(c) Consumed by low-paid workers (d) All of the above

Answer: (d)

10. Veblen good is:

(a) Good of status (b) Consumed by very high-income group

(c) Like diamonds (d) All of the above

Answer: (d)

11. Factor which affects market demand but not individual demand can be:

(a) Number of consumers in the market (b) Age and sex composition of population

(c) Distribution of income (d) All of the above

Answer: (d)

12. Law of demand does not hold in case of:

(a) Emergency (b) Expectation of price rise

(c) Conspicuous goods (d) All of the above

Answer: (d)

13. Contraction of demand is shown by:

(a) Upward movement on the demand curve

(b) Downward movement on the demand curve

(c) Rightward shift of the demand curve

(d) Leftward shift of the demand curve

Answer: (a)

14. Increase in demand is shown by:

(a) Upward movement on the demand curve

(b) Downward movement on the demand curve

(c) Rightward shift of the demand curve

(d) Leftward shift of the demand curve

Answer: (c)

15. Expansion of demand is shown by:

(a) Upward movement on the demand curve

(b) Downward movement on the demand curve

(c) Rightward shift of the demand curve

(d) Leftward shift of the demand curve

Answer: (b)

16. When same units are demanded at a higher price, it shows:

(a) Increase in demand

(b) Expansion in demand

(c) Decrease in demand

(d) Contraction in demand

Answer: (a)

17. When there is fall in the price of complementary good and rise in the price of substitute good, it shows:

(a) Increase in demand

(b) Expansion in demand

(c) Decrease in demand

(d) Contraction in demand

Answer: (a)

18. When income of the consumer falls the impact on price-demand curve of an inferior good is:

(a) Shifts to the right (b) Shifts to the left

(c) There is upward movement along the curve (d) There is downward move

Answer: (a)

19. If with the rise in price of good Y, demand for good X rises, the two goods are:

(a) Substitutes (b) Complements

(c) Not related (d) Jointly demanded

Answer: (a)

20. Any statement about demand for a good is considered complete only when the following is mentioned in it:

(a) Price of the good (b) Quantity of the good

(c) Period of time (d) All of the above

Answer: (d)